

The Testening

Part 2

EEE3094S 2020

Information

- The test is open book.
- The test is out of 50, but there are 55 marks available in the paper. There are five questions.
- Submit your answers in pdf format. They may be typed or handwritten. Sketches may be hand-drawn and scanned, or constructed in a program such as PowerPoint.
- You must explain your reasoning and show your calculations for all questions.
- “Your mom” jokes date back at least 3500 years, with the oldest example found on a Babylonian tablet.
- The world’s oldest ham is 119 years old. It is, allegedly, still edible.

Question 1

The frequency responses of three second-order systems are shown on the Nichols chart in Figure 1.A. They are all stable in open loop.

- A) [3 marks] Are all of the systems stable in closed loop? Explain.
- B) Rank the systems according to how they perform in feedback in each of the following criteria:
- i) [3 marks] Accurate position tracking (most accurate to least accurate)
 - ii) [3 marks] Damping (most damping to least damping)
 - iii) [3 marks] Speed (fastest to slowest)
- C) [3 marks] All three of these systems meet the design requirements for an application, and you must select one to implement as a digital controller. The sampling rate has not been specified. Which one would you recommend? Why?

Total: 15 marks

Q1

A) Yes they are all stable in closed loop. None of the systems pass through the $-180^\circ / 0\text{dB}$ point, which would make them unstable. The plants also stay on the right hand side of the $-180^\circ / 0\text{dB}$ point, which indicates no encirclements of -1 and no unstable closed loop poles. This means that the system is stable in closed loop.

B) i) A and B are most accurate because their closed-loop gain is closest to 0dB . C is least accurate

ii) $\zeta = \text{CL gain} * \text{phase margin} / 100$

A: gain = $-0.55\text{dB} = 0.939$. phase margin = 35° . $\zeta = 0.329$

B: gain = $-0.55\text{dB} = 0.939$. phase margin = 55° . $\zeta = 0.516$

C: gain = $-0.9\text{dB} = 0.902$. phase margin = 80° . $\zeta = 0.716$

Ranked most to least: C,B,A

iii) Consider the frequencies marked on the lines. We know that a Bode plot with a higher natural frequency will respond faster. Therefore the Nichols plots with higher frequencies at higher gains will respond faster. Therefore speed rank is B, C, A

C) Implementing a system with a digital controller will result in the system gaining phase lead. I would recommend C because it has the highest gain margin and will be the least likely to get pushed into the critical point and unstable when it is implemented as a digital controller.

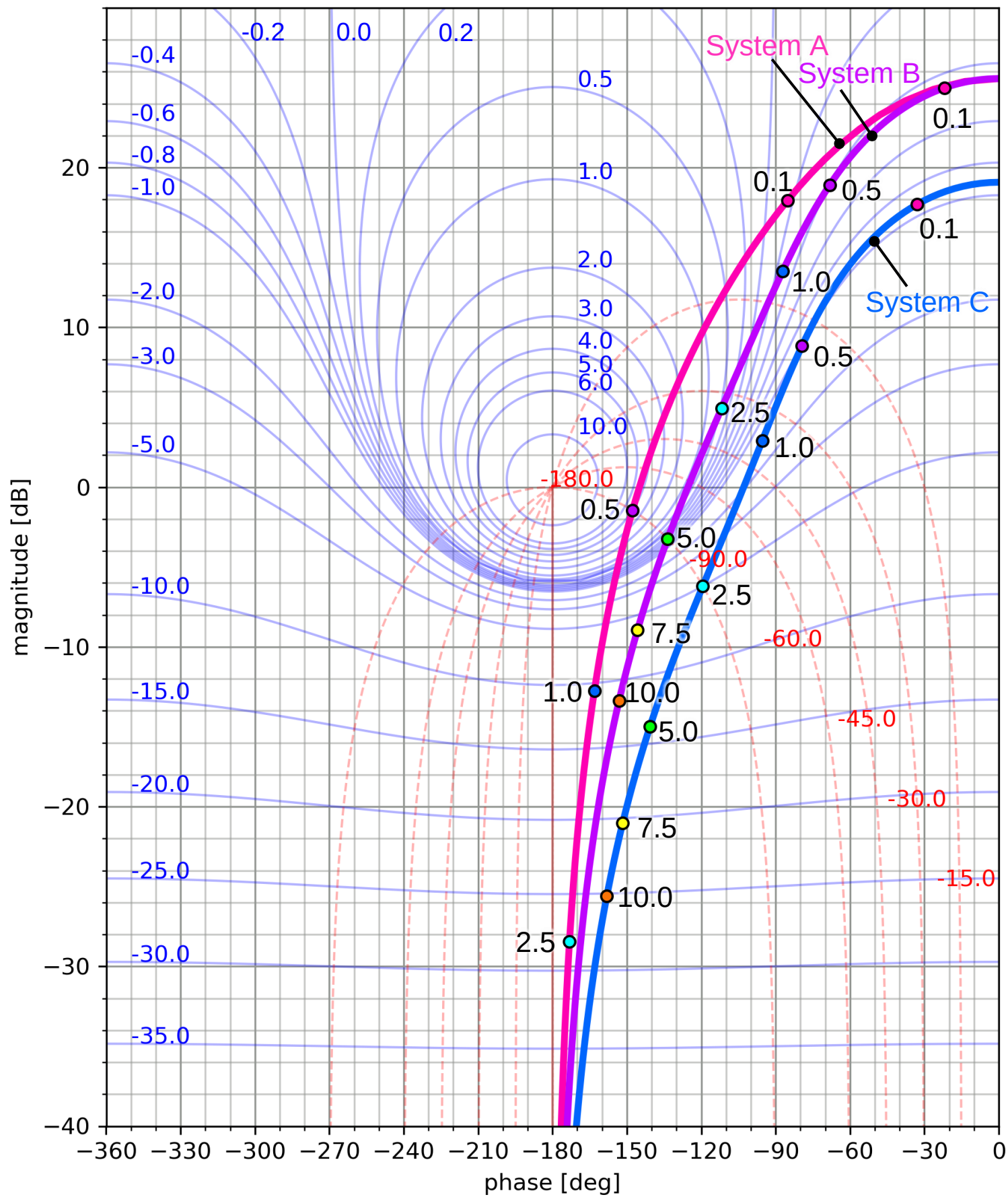


Figure 1.A: Nichols chart comparing systems A, B and C. Labelled frequencies are in radians per second.

Question 2

- A) [6 marks] A first-order system produces the output shown in figure 2.A in response to a unit step. Identify a transfer function for this system. You may model the dead time using a first-order Pade approximation.

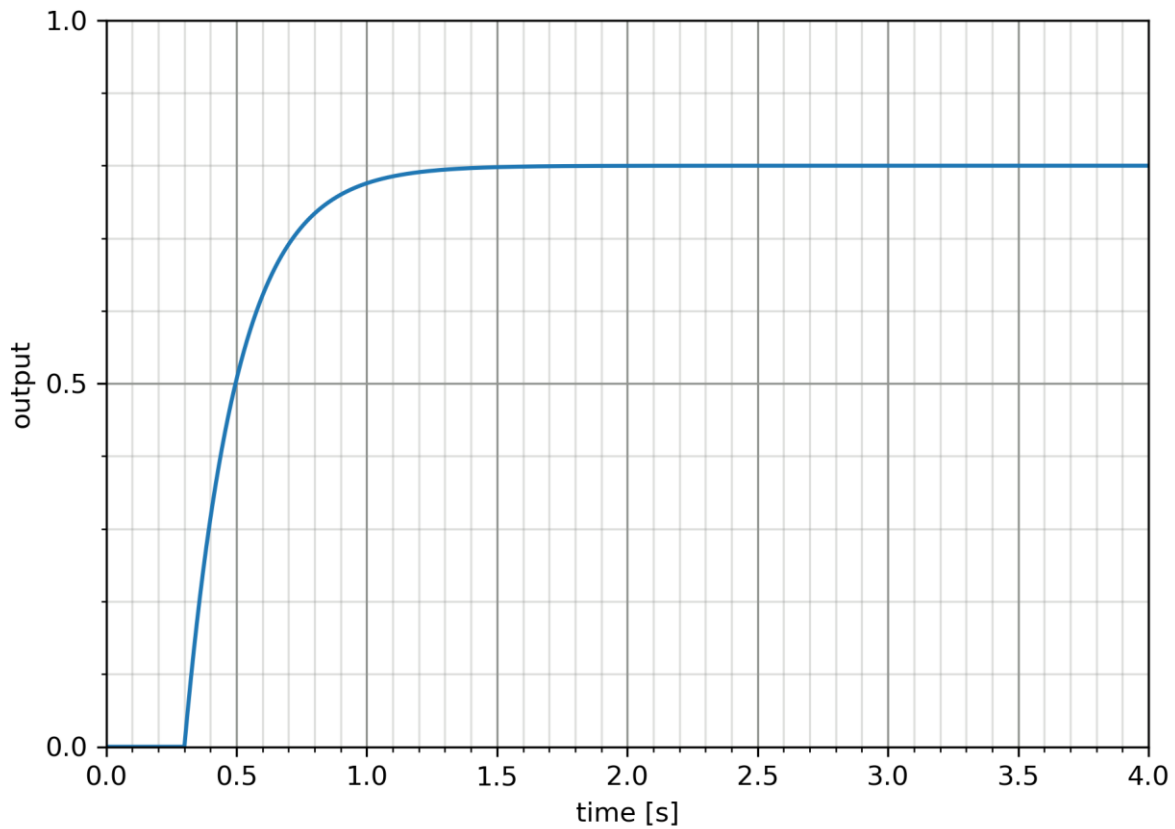


Figure 2.A: unit step response

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- B) [4 marks] The Nyquist plot shown in figure 2.B shows the frequency response of a stable system. Will the system remain stable if it experiences a delay of 0.2 seconds?

Total: 10 marks

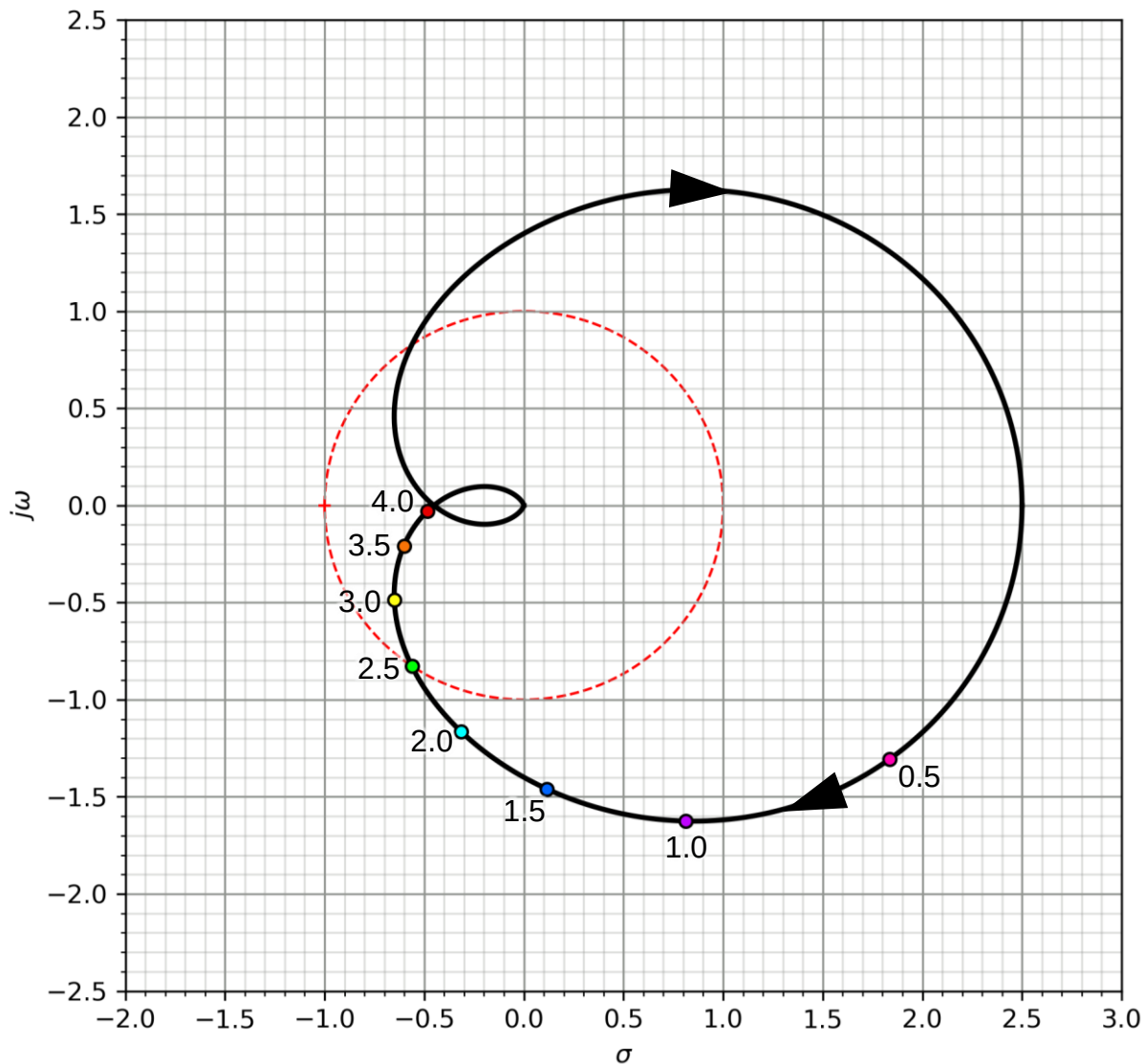


Figure 2.B: Nyquist plot. The labelled frequencies are in radians per second.

- A) Starting by ignoring the dead time and finding the transfer function for the first order system:

$$\text{Gain: } A = \frac{\delta y}{\delta r} = \frac{0.8}{1} = 0.8$$

$$\text{Rise time: } 0.633 * 0.8 = 0.51$$

$$\text{Time taken to reach 0.51: } \tau \cong 0.2s$$

$$\text{Therefore the transfer function of the response is: } \frac{0.8}{0.2s+1}$$

Pade approximation:

A first order Pade approximation is of the form:

$$\frac{1 - \frac{t_d}{2}s}{1 + \frac{t_d}{2}s}$$

In this case it is:

$$\frac{1 - 0.15s}{1 + 0.15s}$$

The final transfer function is:

$$\frac{(0.8 - 0.12s)}{(0.2s + 1)(0.15s + 1)}$$

- B) The effect a delay has on the Nyquist plot is a rotation around the origin. Substituting the provided frequencies into $e^{-0.2j\omega}$ will tell how far the point is rotated. The magnitude will not change.

Frequency	Angle
4	-45
3.5	-40
3	-34
2.5	-28

2.5 is the critical frequency because that is the point that is far enough from the origin to encircle -1.

It's current angle is: $\arctan\left(\frac{0.85}{0.55}\right) = 57^\circ$

Therefore the system will still be stable but it's stability margins will be significantly reduced.

Question 3

A system has poor steady-state tracking accuracy, and it is determined that it can be sufficiently improved by increasing its gain by a factor of 10. Figure 3.A shows the system's frequency response on an inverse Nichols chart.

- A) A controller with the frequency response shown in figure 3.B is suggested for the job.
- [1 mark] Is this a lead compensator or a lag compensator?
 - [4 marks] Would you recommend using this design? If not, suggest how it could be modified to make it suitable.
- B) [4 marks] Would you suggest a proportional controller as an alternative? Explain.

Total: 9 marks

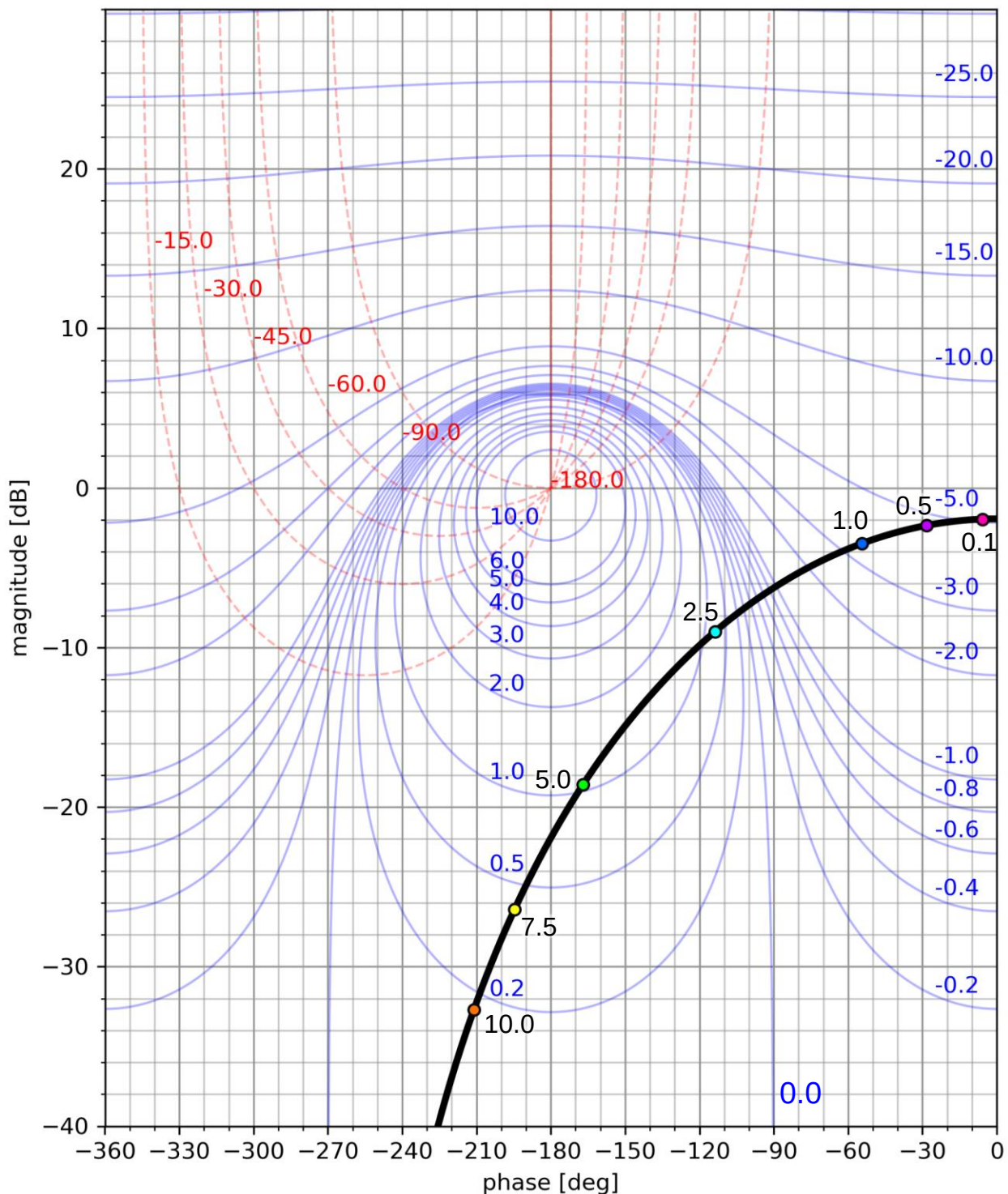


Figure 3.A: Inverse Nichols chart of uncompensated plant

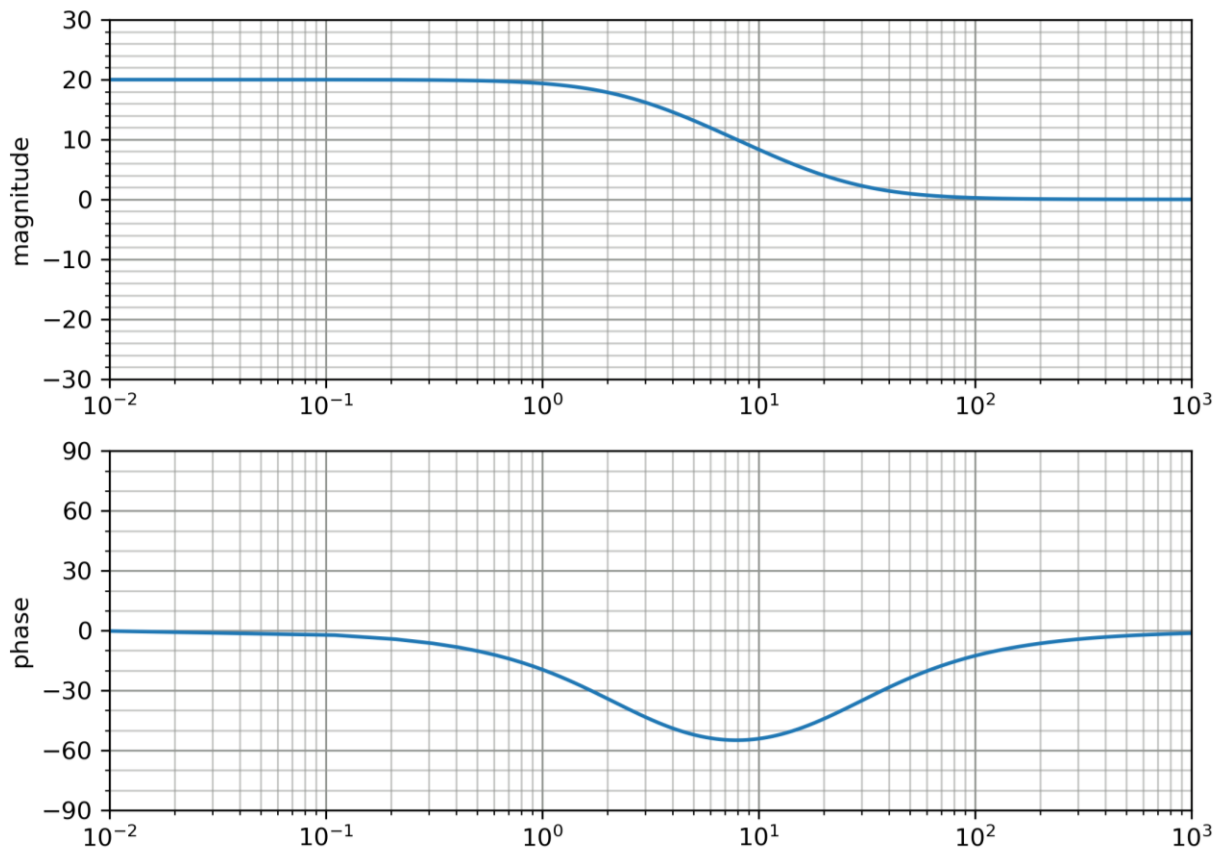


Figure 3.B: Bode plot of suggested controller

A)

- i. The natural frequency of the pole is lower than the natural frequency of the zero. Therefore, it is a lag compensator.
- ii. First find the system's stability margins:
Gain margin: 22dB
Phase margin 60 degrees

To analyse the effect on stability the two frequencies closest to the unstable point will be analysed:

2.5 rad/s

Added gain is approximately 18dB with -40 degrees of phase

New location is 9dB and -75 degrees of phase

5 rad/s

Added gain is approximately 12dB with -50 degrees of phase

New location is -7dB and -130 degrees of phase

The controller would be stable.

Increasing the DC gain by a factor of 10 requires an open loop gain increase of 20dB, which this controller provides.

I would recommend using this design because it will be stable and meet the performance requirements.

TA note – I would encourage students to sketch the new Nichols plot over the old one. I

don't have a printer so I can't just now.

Question 4

Consider the Nichols chart shown in figure 4.A and answer the following TRUE or FALSE questions. Provide a short explanation for each answer. The system in the plot is stable in open and closed loop.

- A) [2 marks] The system will not be destabilized by a delay.
- B) [2 marks] The system will not be destabilized by increasing the gain.
- C) [2 marks] The system can track a step setpoint with $\geq 90\%$ accuracy.

Total: 6 marks

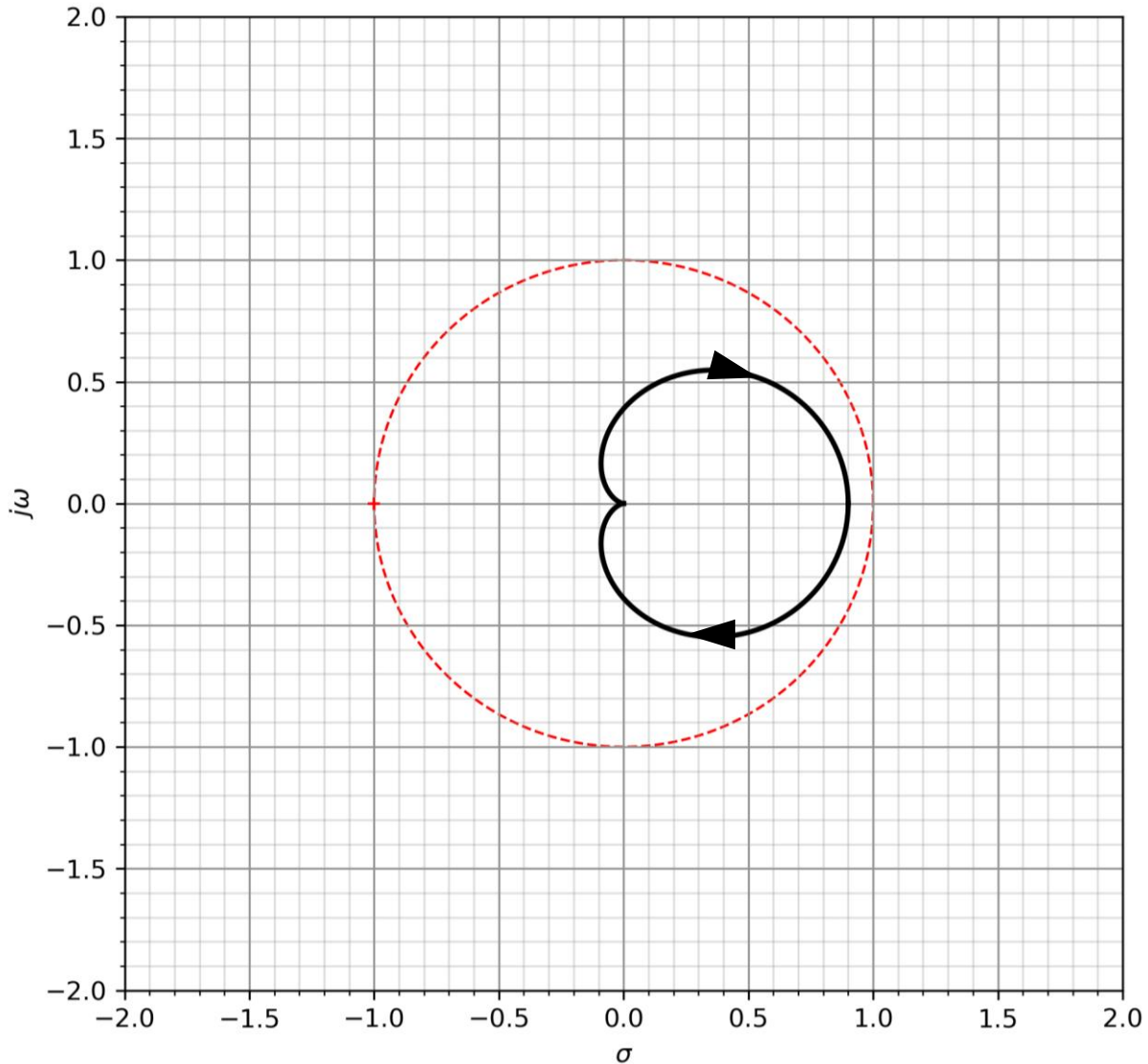


Figure 4.A: Nyquist plot

- A. True, at the current gain. The system is entirely within the -1 circle. Therefore, no amount of rotation will result in instability.
- B. True. The system goes to the origin therefore increasing the gain will not make it encircle -1 because it will always come back to the origin.
- C. False. The DC gain of the system is 0.9. The steady state error is $\frac{1}{1+0.9} = 0.526$. The required error is $< 10\%$

Question 5

A system has open-loop poles at -0.5 and -1. Its frequency response is shown in figure 5.A.

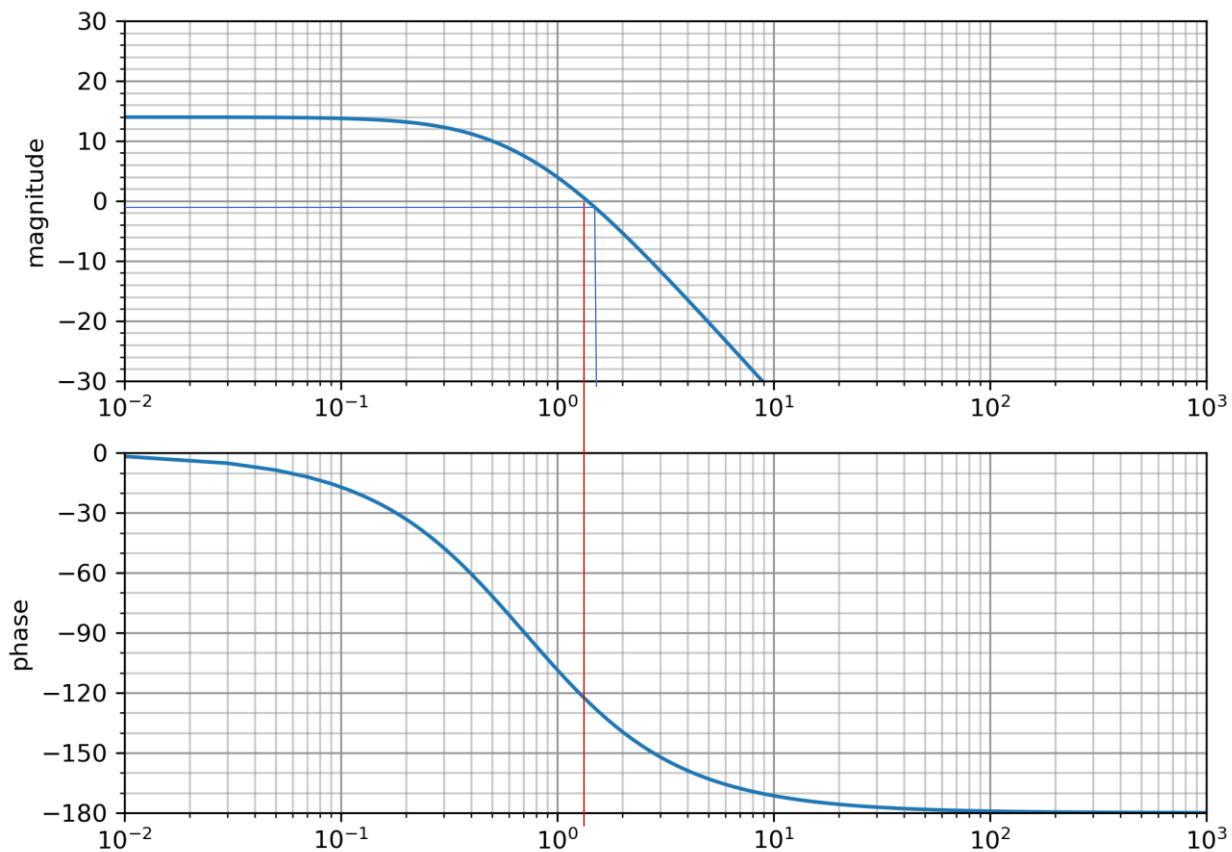
A) Decide whether the system satisfies the following design specifications in closed loop. Justify your answers.

- [2 marks] Phase margin of >60 degrees.
- [2 marks] Percentage overshoot of less than 20%
- [2 marks] Settling time to 2% of its final value in < 2 seconds.

Percentage overshoot: $100e^{-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}}$

B) [9 marks] Design a lead compensator so the closed-loop system will satisfy the specifications in (A). There is no accuracy specification, but you should calculate the position error constant of your design.

Total: 15 marks



Q5:

A) i) The red line shown on the bode plot indicates that it has a phase which is slightly more negative than -120. This means that the phase margin is slightly less than 60 degrees and therefore does not meet the specification of >60 degrees.

ii) Percentage overshoot: $100e^{-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}}$

CL gain = $\frac{5}{6} = 0.833$

$$\zeta = \frac{CL \text{ gain} * \text{phase margin}}{100} = \frac{0.833 * 60}{100} = 0.5$$

$$\text{overshoot} = 100e^{-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}} = 16.3\%$$

This meets the <20% overshoot specification

iii) find ω_n : gain at $\omega_n = CL \text{ gain} - 6 - 20\log \zeta$
 $= 20\log(0.833) - 6 - 20\log(0.5)$
 $= -1.567\text{dB}$

Using blue line on bode chart, $\omega_n = 1.6\text{rad/s}$

$$t_{2\%} = \frac{4}{\zeta\omega_n} = 5 \text{ seconds. This violates the } t_{2\%} < 2 \text{ seconds requirements}$$

- B) Student's will answer this in varying ways. Award marks for correct steps and good reasoning. They must analyse the closed loop error.